

HexRuinx

Cisco Certificate Network Associate 200-301

v1.1

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Introduction 3

Resources 3

The BASIC OverView 3

Memory 4

Internal Memory 4

Read Only Memory (ROM) 4

Types of ROM..... 4

Erasable Programmable Read Only Memory (EPROM) 4

Electrically Erased Programmable Read Only Memory (EEPROM)..... 4

Types of Random Access Memory 4

Static Random Access Memory 4

Dynamic Random Access Memory (DRAM)..... 4

Type of DRAM..... 5

Synchronous Dynamic Random Access Memory 5

Central Processing Unit (CPU)..... 5

Network Interface Controller (NIC) 6

Graphics Processing Unit (GPU) 6

External Memory 6

Hard Drive 6

Type of Hard Drive 6

Hard Disk Drive (HDD) 6

Solid State Drive (SSD)..... 6

Low-Power Double Data Rate..... 6

CABLES 6

PATA Cables 7

SATA Cables 7

M2 7

Non-Volatile Memory Express (NVME)..... 7

Network Topology & Components..... 7

Network Topology.....	8
Point to Point Topology (P2P).....	8
Bayonet Neil Concelman (BNC Connector)	8
Bus Topology	8
Ring Topology	9
Star Topology.....	10
Hybrid Topology	11
Mesh Topology.....	12
Half Mesh Topology	13
Full Mesh Topology.....	13
Network Cables.....	14
Coaxial Cables.....	15
Ethernet Cables	15
Types of Cables.....	15
Unshielded Twisted Port	15
Shieled Twisted Pair	15
Ethernet Cable Types.....	15
Device Pin Connections:.....	17
Cable Standard:	17
Fiber Optic Cable:	18
Small Form-Factor Pluggable (SFP).....	18
Long && Short Range SFP.....	18
MDIX Technology:.....	19
Internet Protocol	19
Medial Access Control.....	20
IP Address Version	20
Types of IP Address	21
Network Address Translation.....	21
Enumeratrix IP Address.....	22

Subnetting	23
Example:	23
Formula for Subnet Mask Calculations	24
Network Bits and Host Bits for Each IP Class	24
Subnet Address.....	25
Formula for Subnet Mask.....	26
Fixed Length Subnet Mask (FLSM).....	27
Variable Length Subnet Mask (VLSM)	27
Format of An IPv4 Packets.....	27
Internet Protocol V6.....	29
Internet Protocol v4 VS v6	31
OSI MODEL LAYER.....	32
International Standard Organization (ISO)	32
Open System InterConnection (OSI)	32
OSI Model Layers	32
Application Layer	33
Presentation Layer	33
Session Layer.....	33
Transport Layer	33
Network Layer.....	34
Data Link Layer	34
Physical Layer.....	34
OSI Topology.....	34
OSI Layer Vs Attacker Layer.....	35
TCP / IP Protocol	36
TCP / IP Layers	37
TCP/IP Network Topology.....	37
Network Protocol And Service	38

Introduction

Welcome! I am a Security Engineer working in the field of Security. With a passion for all things related to the industry, I created this GITBOOK as a way to share the information and insights I have gained throughout my career. I hope it serves as a helpful resource for others in the field and as a central for the references I frequently use.

Please note that the opinions expressed on this Gitbook and my connected social media profiles are solely my own and do not represent the views or positions of my employer. Mentions of specific assets should not be interpreted as endorsements of any position, product, or service. Additionally, this website does not generate any income for me. I would greatly appreciate receiving information about any errors noticed by the readers so that they may be corrected as soon as possible in the future.

Contact me through any social media platform. However, if you would like to support my efforts, you can send me a message to buy me a coffee. Thank you for your understanding

Resources

MEGA: <https://mega.nz/folder/NL1RkZDA>

Gitbook : <https://hexruinx.gitbook.io/ccna>

Download Document:

The BASIC OverView

Memory

Memory is used to Store DATA.

Internal Memory

Read Only Memory (ROM)

A Small Chip Which Designs to complete TASK.

A Washing Machine where the timer runs out of time and the Machine Stops this is because of ROM which is configured to do Permanent Task.

A Phone is configured to BOOT with LOGO of the manufacture Company every PowerUp

Types of ROM

Erasable Programmable Read Only Memory (EPROM)

The ROM which can be reprogrammed or Erasable.

Electrically Erased Programmable Read Only Memory (EEPROM)

The ROM which can be reprogrammed or Erasable via Electricity.

Random Access Memory (RAM)

The temporary task is performed by RAM. This information will be lost after a Power boot.

Types of Random Access Memory

Static Random Access Memory

The information which is stored in SRAM will not be deleted until by the user intervention.

Dynamic Random Access Memory (DRAM)

The Information stored in DRAM is changeable at high speed.

Type of DRAM

Synchronous Dynamic Random Access Memory

The OLD RAM could not sync with systems which results in no logs being generated.

So, the SDRAM provides both sync and logs generation about how long the application service is running. The SDRAM speed is '**133 MHZ**'

Rambus Dynamic Random Access Memory (RDRAM)

It was created at Private Sector, which handles heavy application/tasks in 1990s where the speed is '**1GHZ**'

Double Data Rate Random Access Memory (DDRAM)

It's the updated version of DRAMs which is faster and compatible and created by **IBM**. The versions are DDR1 to DDR6 and more to go.

To Check the RAM OEM, navigate to Computer,>> Right click and click on manage. Then Device Manager.

Model	Details
DDR1	it can be up to 2GB RAM.
LIDDR1	Date: 1998 Voltage: 2.5/2.6 Bandwidth Speed: 1600
DDR2	2 x 2 equals 4 GB of RAM. The date is 2003, the voltage is 1.8, and the bandwidth speed is 2133.
DDR3	2 x 2 x 2 equals 8 GB RAM. The date is 2007, voltage is 1.5/1.35, and the bandwidth speed is 2666.
DDR4	Date: 2014 Voltage: 1.2/1.05 Bandwidth Speed: 3200
DDR6	2 x 2 x 2 x 2 x 2 x 2 = 64gb ram.

DIMM

The laptops have different number of slots where RAM can be inserted. These slots are called DIMM slots which can be viewed in **TASK MANAGER**.

Central Processing Unit (CPU)

The CPU is the Brain of Computer which commands and manages Tasks/Programmed.

CPU Cores are Single, Dual, Quad, Octa, Hexa, Deca and more to go

Network Interface Controller (NIC)

It's the network adapter hardware which connects computer to LAN.

Graphics Processing Unit (GPU)

It's the updated version of RAM which is a power horse for running the Application.

External Memory

Hard Drive

It stores all the data, created by IBM with a capacity of 5MB.

Type of Hard Drive

Hard Disk Drive (HDD)

It's a Disk where the data is stored and if the disk is damaged the data is lost.

Solid State Drive (SSD)

It's a green board with nano participants disk which stored the data where the data is recoverable.

This Hard drive is connected to SATA drive (PATA >><< SATA) Pata was used before SATA.

Low-Power Double Data Rate

It's used in mobile devices to support high-bandwidth data transfers while using less power.

CABLES

PATA Cables

It was first cable in industry to transfer data from **Hard Drive** to ram (up to 1GB)

SATA Cables

It was invented after PATA Cable, it is also used to share data form HDD to RAM (like up to 2GB)

M2

It is also a data cable and more efficient than SATA Cable (up to 4GB).

Non-Volatile Memory Express (NVME)

Its lasted cable is used to transfer data from Hard Drive to RAM. And it was first cable in industry to transfer data from Hard Drive to RAM (Random Access Memory) RAM (up to 1GB)

Network Topology & Components

A network topology is the physical and logical arrangement of nodes and connections in a network. Nodes include devices like switches, routers, and software with switch and router features

Network Topology

The Topology Means the arrangements or layouts where the Network Topology means Network Diagram or layout of Network.

Point to Point Topology (P2P)

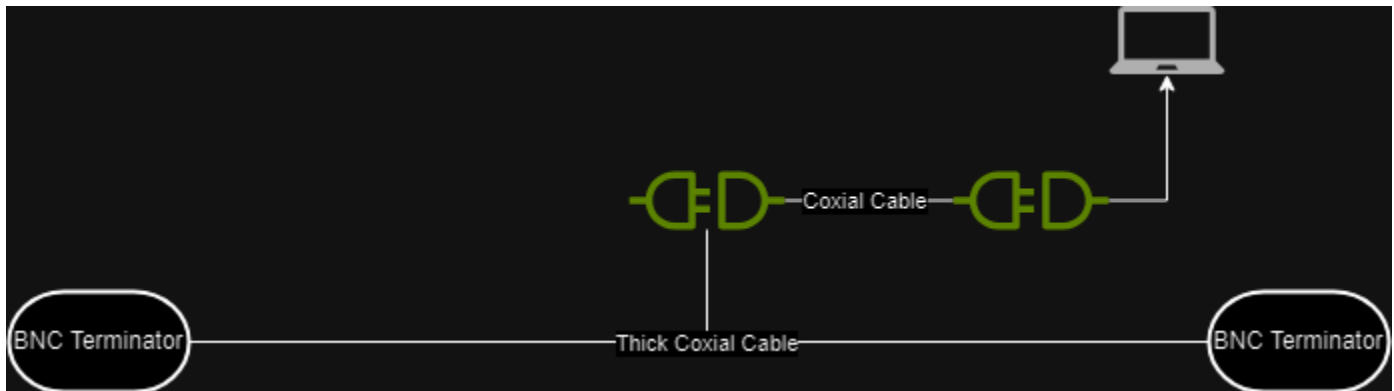
Point to point means the two computers are connected directly to each other.

Bayonet Neil Concelman (BNC Connector)

The BNC Connector is used with coaxial cable to connect devices. Where data transfer in single directions at the single time.

Bus Topology

It's the first which used more than two computers connected using single coaxial cable.



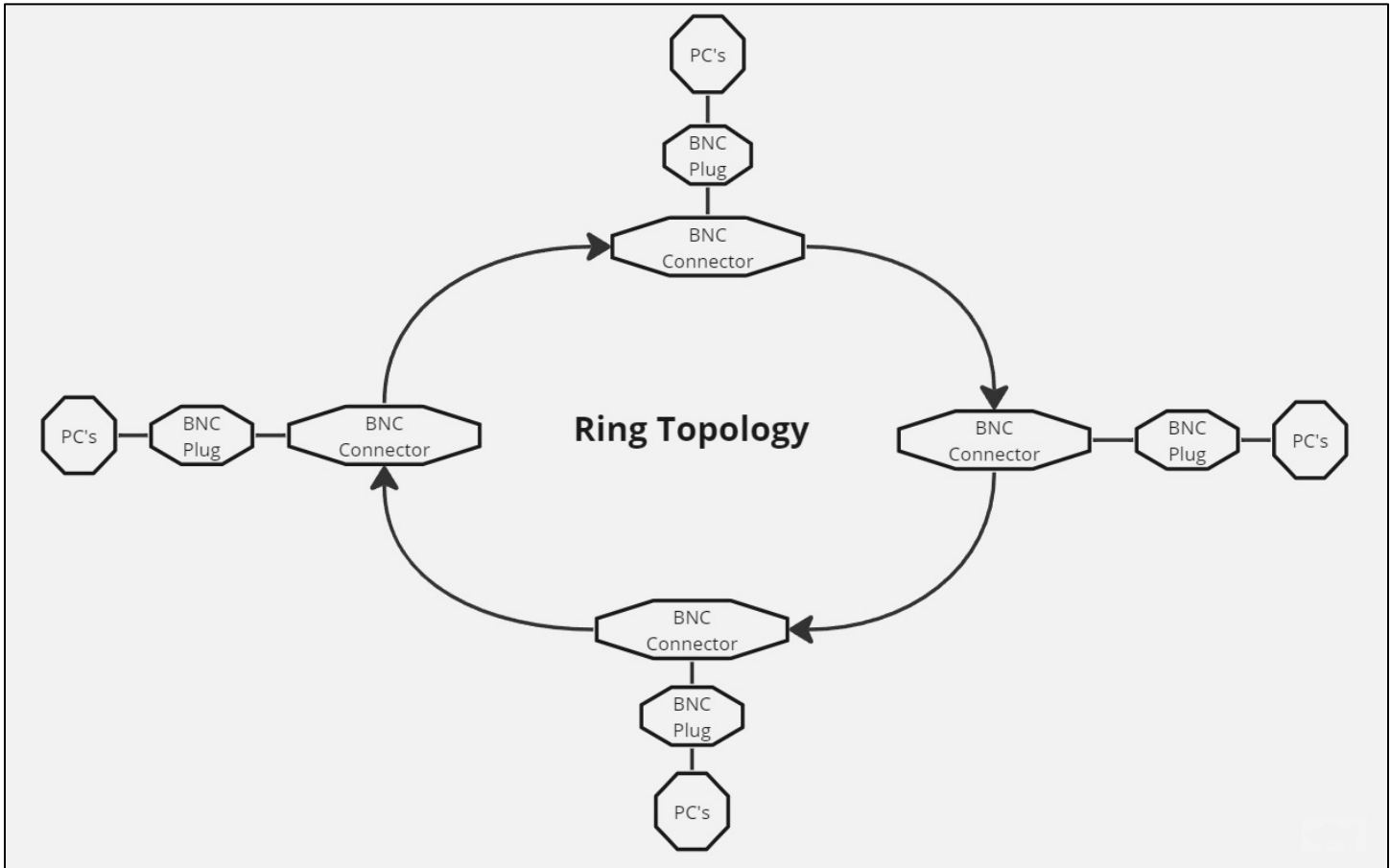
Pros	Cons
Bus topology is cheap,	Files doesn't have secret (it broadcasts data to all)
Use a single coaxial cable which connects all, including nodes.	There is not backup when the cable is damage
	A single point of network failure

Ring Topology

Every device is connected to exactly two other devices, forming a ring for communication.

Data travels in one direction (or sometimes both, in a dual ring).

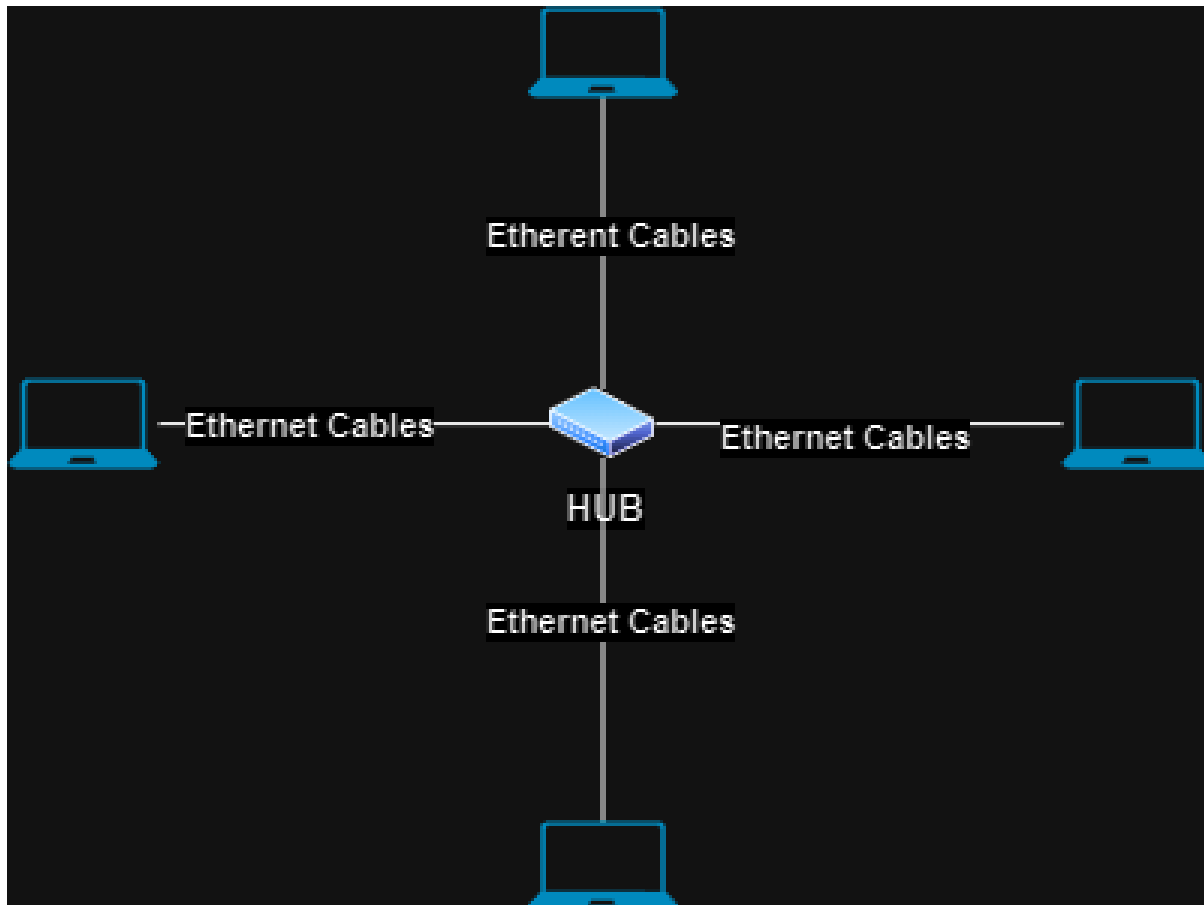
The failure of a single device can affect the entire network unless it's a dual ring.



Pros	Cons
Its has a backup ways to destination in networks	Its doesn't have BNC Terminator . & ts has struggle to add new Pc's because ring Wire has to be the connected wires, It has BNC connector.

Star Topology

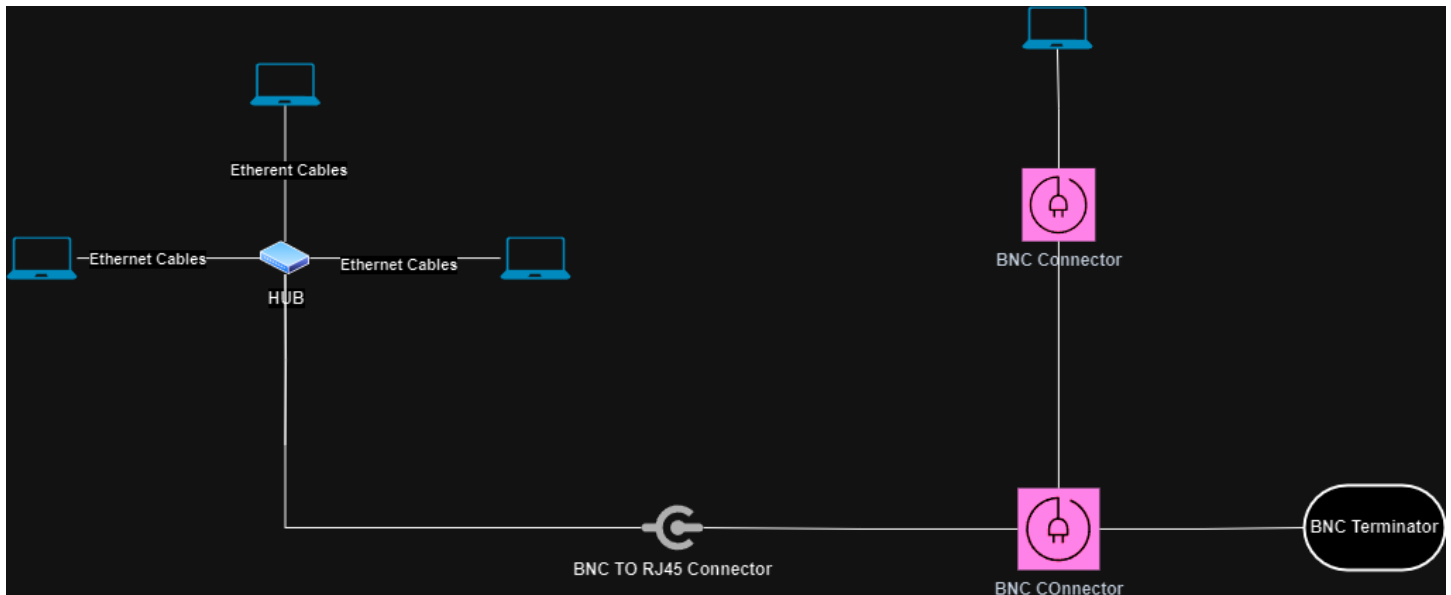
All the devices are connected to a Central HUB or Switches.



Pros	Cons
If one PC is down & cable is physically damaged, the others are not affected.	A hub or switch is a single point of failure for the network

Hybrid Topology

A Combination of two or more different Types of topologies



Pros	Cons
Flexible And Scabble for Large and Complex network	Can be more expensive and complex to set up and manger

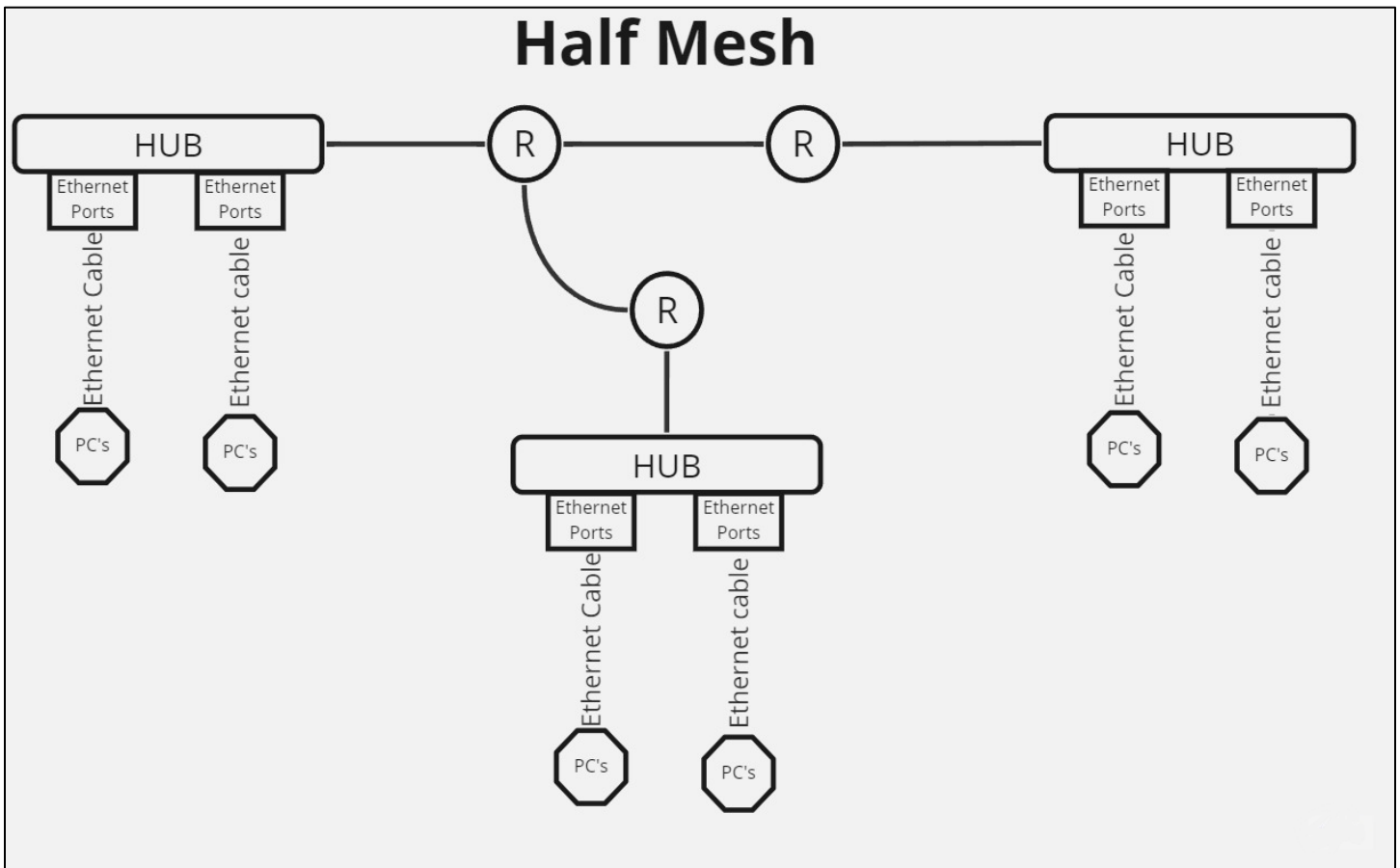
Mesh Topology

Every device is connected to every other devices.

Pros	Cons
High Redundancy and reliability Multiply paths for data flow	Expensive and complex to install and maintain

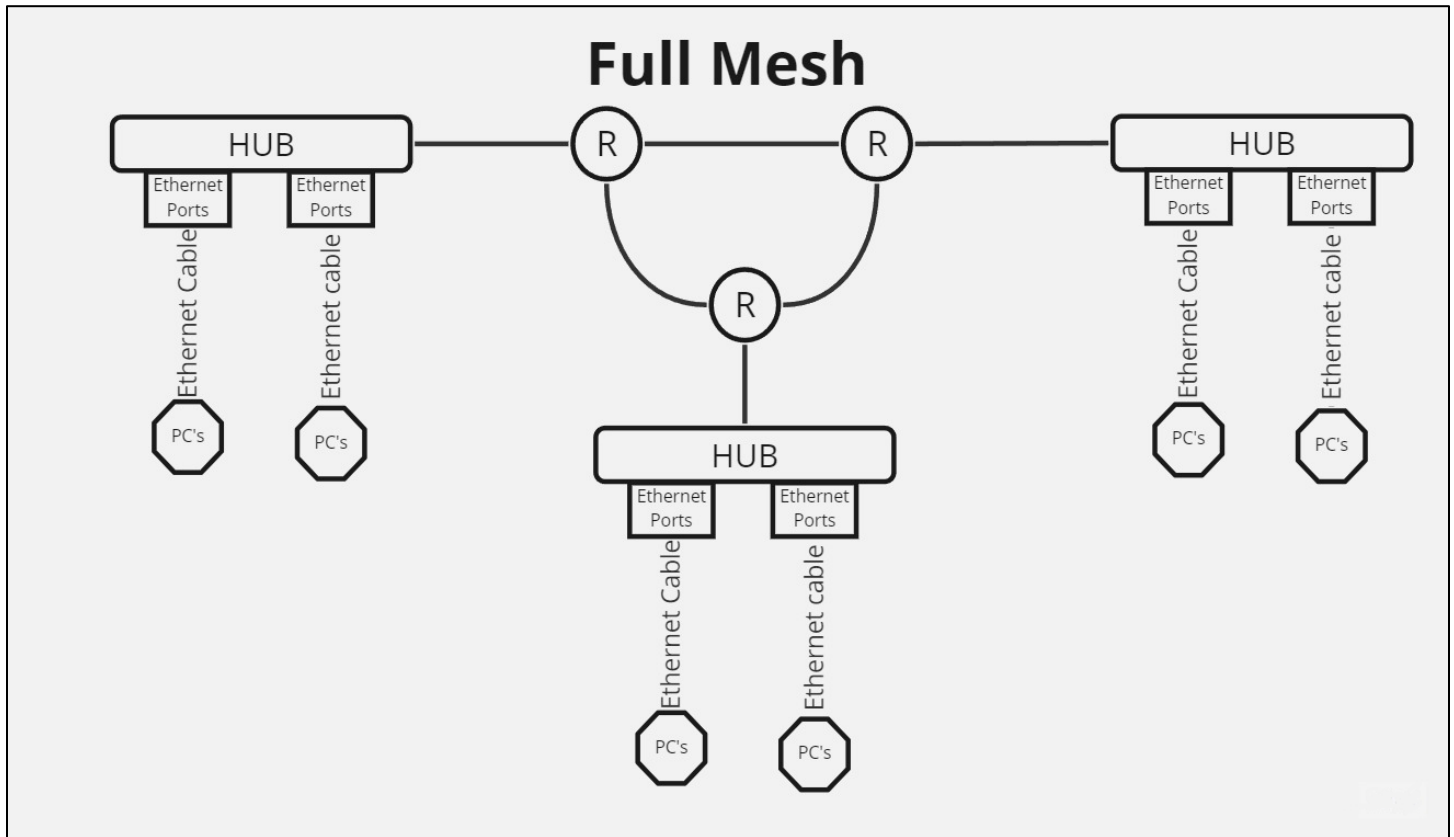
Half Mesh Topology

Only some devices are connected to Multiple others Meaning only one way to the destination.



Full Mesh Topology

Every device has multiply ways to the destination devices.



Network Cables

Network Cables are the connector hardware used to connect computers or network devices.

Coaxial Cables

Its all started with the Coaxial Cables where its made of Plastic and copper wires which transmit the signals in electrical forms.

Ethernet Cables

Ethernet are the new technologies now a days used everyone. its standard of IEEE (Institute of Electrical and Electronic Engineers) 802.3 in 1983.

Speed	Common Name	IEEE Standard	Informal Name	Max Lengrh
10 MBPS	Ethernet	802.3i	10 Base -t	100m
100 MBPS	FastEthernet	802.3u	100Base -t	100m
1GBPS	GigabitEthernet	802.3ab	1000Base -t	100m
10 GBPS	TenGigabitEthernet	802.3an	10G Base -t	100m

Types of Cables

Unshielded Twisted Port

Its a 4 pair cables (total 8 wires) are twisted together which help eliminate electromagnetic interference, Where these 8 wires are connected to 8 Pins present in the RJ-45 connector. cables which don't have aluminum shield and vulnerable to electrical disturbance.

Shieled Twisted Pair

Its the same cables as UTP but more advantages like aluminum shield.

Ethernet Cable Types

FastEthernet And Ethernet (10/100 Mbps)

These cables use 4 wires.

Pin 1 and Pin 2 are used for transmitting data from the PC to the switch.

Pin 3 and Pin 6 are used for receiving data from the switch to the PC

Gigabit Ethernet and 10 Gigabit Ethernet:

These cables use 8 wires.

Provide higher data rates and require more wires for transmission

Connection Between PC and Switch (Straight Through Cable):

Transmit Data (PC) → Receive Data (Switch):

Pin 1 (Transmit) on PC to Pin 1 (Receive) on Switch.

Pin 2 (Transmit) on PC to Pin 2 (Receive) on Switch.

Receive Data (PC) → Transmit Data (Switch):

Pin 3 (Receive) on PC to Pin 3 (Transmit) on Switch.

Pin 6 (Receive) on PC to Pin 6 (Transmit) on Switch.

Connection Between Switch and Router (Straight Through Cable):

Similar to PC to Switch connection, as the router behaves like a PC in this context.

Crossover Cable (Same Device to Same Device):

When connecting devices of the same type (e.g., router to router), a crossover cable is used.

Pin 1 (Transmit) on one end to Pin 3 (Receive) on the other end.

Pin 2 (Transmit) on one end to Pin 6 (Receive) on the other end.

Pin 3 (Receive) on one end to Pin 1 (Transmit) on the other end.

Pin 6 (Receive) on one end to Pin 2 (Transmit) on the other end.

This information outlines how Ethernet cables are used to establish connections between various network devices, covering both straight-through and crossover cable configurations.

Device Pin Connections:

Device	Transmit	Receive
Router	1 and 2	3 and 6
Firewall	1 and 2	3 and 6
PC	1 and 2	3 and 6
Switch	3 and 6	1 and 2

- For gigabit Ethernet and 10 gig, for straight connections, all cables are connected straight.
- New devices aren't affected by this because of AUTO MDI-X feature. The device detects which pins the device uses to transmit data and accordingly adjusts their own pins.

Cable Standard:

TIA/EIA 568A Wiring	TIA/EIA 568B Wiring
White Green	White Orange
Green	Orange
White Orange	White Green
Blue	Blue
White Blue	White Blue
Orange	Green
White Brown	White Brown
Brown	Brown

- Mostly Used Wiring is TIA/EIA 568B Wiring.
- If the same standard cable is applied on both sides, it's called a Straight Cable.
- If both are different, then it's called a crossover.
- Example: Switch to Hub, Server to PC are Crossover.
- ORANGE & GREEN / BOTH WHITE are the data flow wires.
- BLUE & BROWN / WHITE are for supplying electric power to devices.

Fiber Optic Cable:

There are two types of fiber cables:

1. Single mode: Core is smaller than multi-mode.
2. Multi-mode: Core is wider than single mode, allows multiple modes of light waves to enter the glass.

Informal Name	IEEE Standard	Speed	Cable Type	Max Length
1000base-lx	802.3z	1gbps	Multi-mode or single mode	550m (MM) -
5km (SM) 10gbase-sr	802.3ae	10gbps	Multi-mode	400m
10gbase-lr	802.3ae	10gbps	Single mode	10km
10gbase-er	802.3ae	10gbps	Single mode	30km

Small Form-Factor Pluggable (SFP)

Small Form-factor Pluggable (SFP) module is a compact, hot-swappable transceiver used for both telecommunication and data communication applications.



Long & Short Range SFP



Long Range SFP is used for long range communication, Both Side should be equal.

Short Range SFP is used for Short range communication, Both Side should be equal.

MDIX Technology:

If accidentally applying the wrong standard of cable, this feature automatically changes the energy to both connections.

This allows them to cooperate; NIC cards support MDIX Technology.

Internet Protocol

It's a set of rules governing the format of data sent over the internet or local network.

Which was created by ICANN also known as **IANA** (Internet Assigned Number Authority)

An IP address is a unique address that identifies a device on the internet or a local network. It can be static (permanent) or dynamic (temporary).

IANA (Internet Assigned Numbers Authority) is responsible for coordinating global IP address allocation by distributing IP address blocks to regional internet registries (RIRs).

Media Access Control

A MAC (Media Access Control) address is a unique identifier assigned to a network interface controller (NIC) for use as a network address within a network segment.

Devices need to find each other on a **network** to establish communications. These devices use **unique addresses**, or **identifiers**, to locate each other. These addresses ensure that communication happens with the right device. These are called **IP and MAC addresses**.

Think of a MAC address as a permanent identifier like a serial number on a device, whereas an IP address is like a temporary postal address assigned to the device when it connects to a network.

Example of IP and MAC: A company gives you an **ID card** called an **IP address**. **the personal identity is (AADHAR) called a MAC address**. The company hires you based on your MAC address, and then you get an IP address.

IP Address Version

IANA has IP Address its product for them so they use to launch it with version.

- IP Address has **IPv4 and IPv6 versions all the other version was unlaunched**.
- IPv4 is made of 4 blocks, each block holds 8 bits.
- IPv4 only uses decimal numbers.
- IPv6 was introduced due to IPv4 shortages.
- IPv6 is made of 8 blocks, each block holds 16 bits.
- IPv6 uses hexadecimal numbers.

IP Address Classes

Class	Range	Details
Class A	1.0.0.0 - 126.255.255.255	Used By ISP & its Public IP
	127.0.0.1	NIC Card check (Loopback)
Class B	128.0.0.0 - 191.255.255.255	Used By ISP & its Public IP
Class C	192.0.0.0 - 223.255.255.255	Used By ISP & its Public IP
Class D	224.0.0.0 - 239.255.255.255	Network Device Communication
Class E	240.0.0.0 - 255.255.255.255	Research Centers

- Address range: **0-255.0-255.0-255.0-255.**
- Dynamic combination.
- First & last part should not be zero.
- **ISPs (Internet Service Provider)** can use Class **A, B, or C** which is our internet wale Bhaiya.
- Last number zero indicates network.
- One IP can maintain 65,535 devices (connection & port numbers).

Types of IP Address

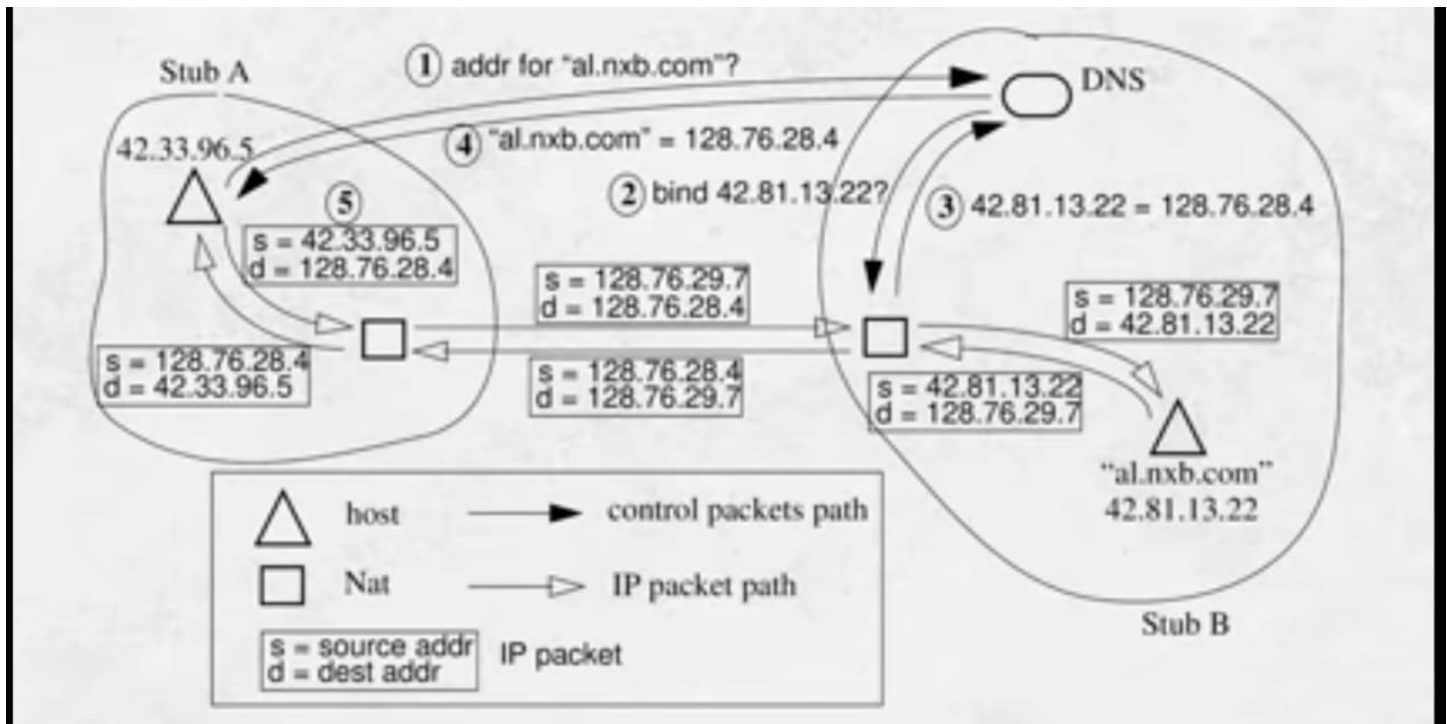
- **Public:** Used on the internet. No simultaneous identical IP and port for two machines.
- **Private:** Used in LAN. Different LANs can have the same IPs. Machines in LAN can't access the web

Network Address Translation

Converting private IP addresses into public IP addresses. Private IP addresses work under LAN but need NAT to access WAN.

Due to increased electronic devices, IP address shortage led IANA to introduce Private Addresses with NAT.

PIX Engineer Created NAT Which was Network Translation Inc. & Cisco Buy PIX and come up with PIX ASA Adaptive Security Appliance



Class	Range	Hosts
Class A	10.0.0.0 - 10.255.255.255	16,777,216 IP addresses
Class B	172.16.0.0 - 172.32.255.255	1,948,576 IP addresses
Class C	192.168.0.0 - 192.168.255.255	65,536 IP addresses

These are private addresses for local network use. They can't access the internet directly. **Private IPs** are converted to public for internet access.

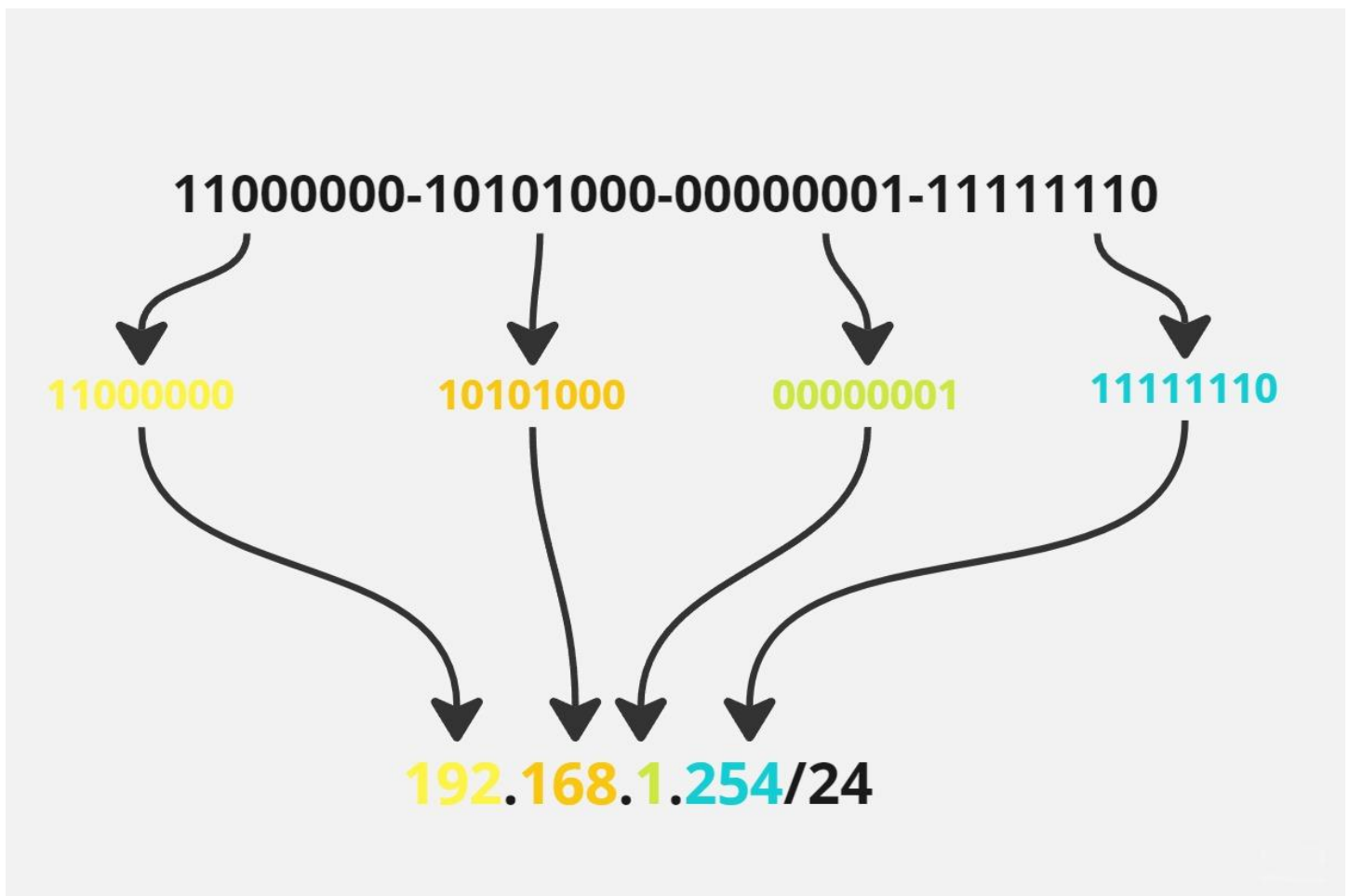
Enumeratrix IP Address

- IPv4 is divided into 4 parts, **0-255.0-255.0-255.0-255.**
- Each part has **8 bits**.
- IPv4 has a total of **32 bits**.
- Each part ranges from **0-255**.
- IP can't start or end with **0** but can have **0** in other places.
- IP ending with **0** represents the network.

Subnetting

- Subnetting divides big networks.
- Subnetting controls IP addresses.
- Subnets are the bits within an IP address.

Example: 192.168.1.0 has a subnet mask of 255.255.255.0, which provides 65,536 IP addresses as free addresses in LANs.



- **Subnetting** involves dividing subnets.
- Network bits are fixed and cannot be changed.
- Host bits are changeable

Example:

- **Class A:** 10.0.0.0
 - Network bits: 10
 - Host bits: 0.0.0. (32 - 8 = 24 host bits)
 - Class A 10.0.0.0 has 8 network bits and 24 host bits.
- **Class B:** 172.16.0.0
 - Network bits: 172.16
 - Host bits: 0.0. (32 - 16 = 16 host bits)
 - Class B 172.16.0.0 has 16 network bits and 16 host bits.
- **Class C:** 192.168.1.0
 - Network bits: 192.168.1
 - Host bits: 0. (32 - 24 = 8 host bits)
 - Class C 192.168.1.0 has 24 network bits and 8 host bits.

Formula for Subnet Mask Calculations

N	2 ⁿ
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128

$$> N = 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 = 255$$

Network Bits and Host Bits for Each IP Class**Class A**

10.	0.	0.	.0
128+64+32+16+8+4+2+1	128,64,32,16,8,4,2,1	128,64,32,16,8,4,2,1	128,64,32,16,8,4,2,1
255	0	0	0

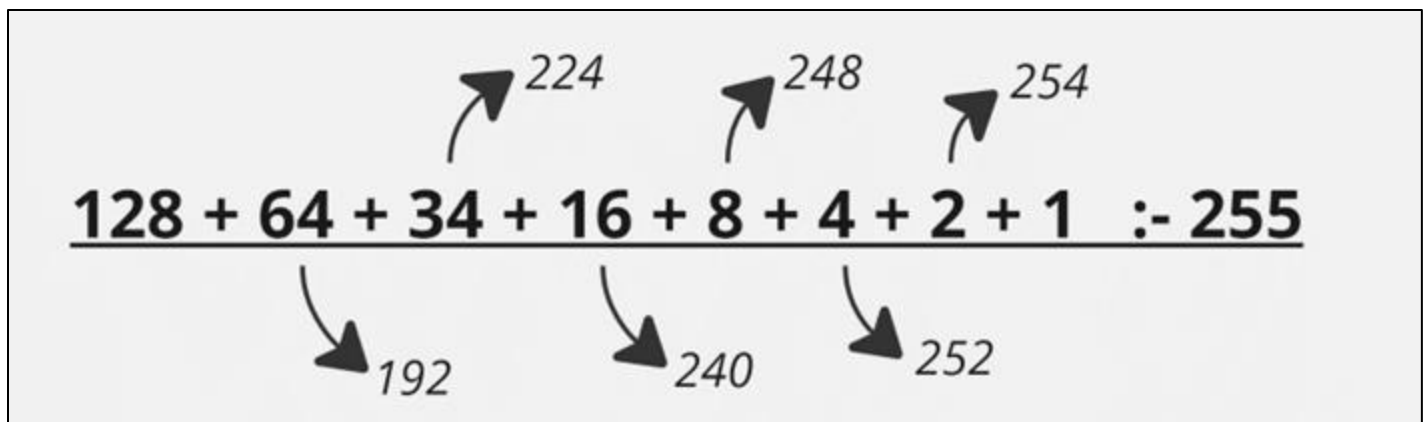
Network Bits	Host Bits	Host Bits	Host Bits
--------------	-----------	-----------	-----------

Class B

172.	16.	0.	.0
128+64+32+16+8+4+2+1	128+64+32+16+8+4+2+1	128,64,32,16,8,4,2,1	128,64,32,16,8,4,2,1
255	255	0	0
Network Bits	Network Bits	Host Bits	Host Bits

Class C

192	168.	1.	.0
128+64+32+16+8+4+2+1	128+64+32+16+8+4+2+1	128+64+32+16+8+4+2+1	128,64,32,16,8,4,2,1
255	255	255	0
Network Bits	Network Bits	Network Bits	Host Bits



Subnet Address

Number/	Range	Details
/8	255.0.0.0	Default of Class A Network ID
/9	255.128.0.0	
/10	255.192.0.0	
/11	255.224.0.0	
/12	255.240.0.0	
/13	255.248.0.0	
/14	255.252.0.0	
/15	255.254.0.0	
/16	255.255.0.0	Default of Class B Network ID
/17	255.255.128.0	
/18	255.255.192.0	
/19	255.255.224.0	
/20	255.255.240.0	
/21	255.255.248.0	
/22	255.255.252.0	
/23	255.255.254.0	
/24	255.255.255.0	Default of Class C Network ID
/25	255.255.255.128	
/26	255.255.255.192	
/27	255.255.255.224	
/28	255.255.255.240	
/29	255.255.255.248	
/30	255.255.255.252	
/31	255.255.255.254	
/32	255.255.255.255	Default Universal Broadcast ID

Formula for Subnet Mask

Class A 10.0.0.0 has 8 Network Bits and 24 Host Bits. Host Bits (24) – Total Subnet = Converted Host Bits. Network Bits (8) + Converted Host Bits = Subnet Mask.

Class B 172.16.0.0 has 16 Network Bits and 16 Host Bits. Host Bits (16) – Total Subnet = Converted Host Bits. Network Bits (16) + Converted Host Bits = Subnet Mask.

Class C 192.0.0.0 has 24 Network Bits and 8 Host Bits. Host Bits (8) – Total Subnet = Converted Host Bits. Network Bits (24) + Converted Host Bits = Subnet Mask.

Fixed Length Subnet Mask (FLSM)

- FLSM stands for Fixed Length Subnet Mask.
- Its subnet size can't be changed; also known as Fixed Size.
- FLSM means a large IP can be divided into an equal number of smaller networks. For example, the network can be divided into 4 subnets, each with 40 IPs.

Variable Length Subnet Mask (VLSM)

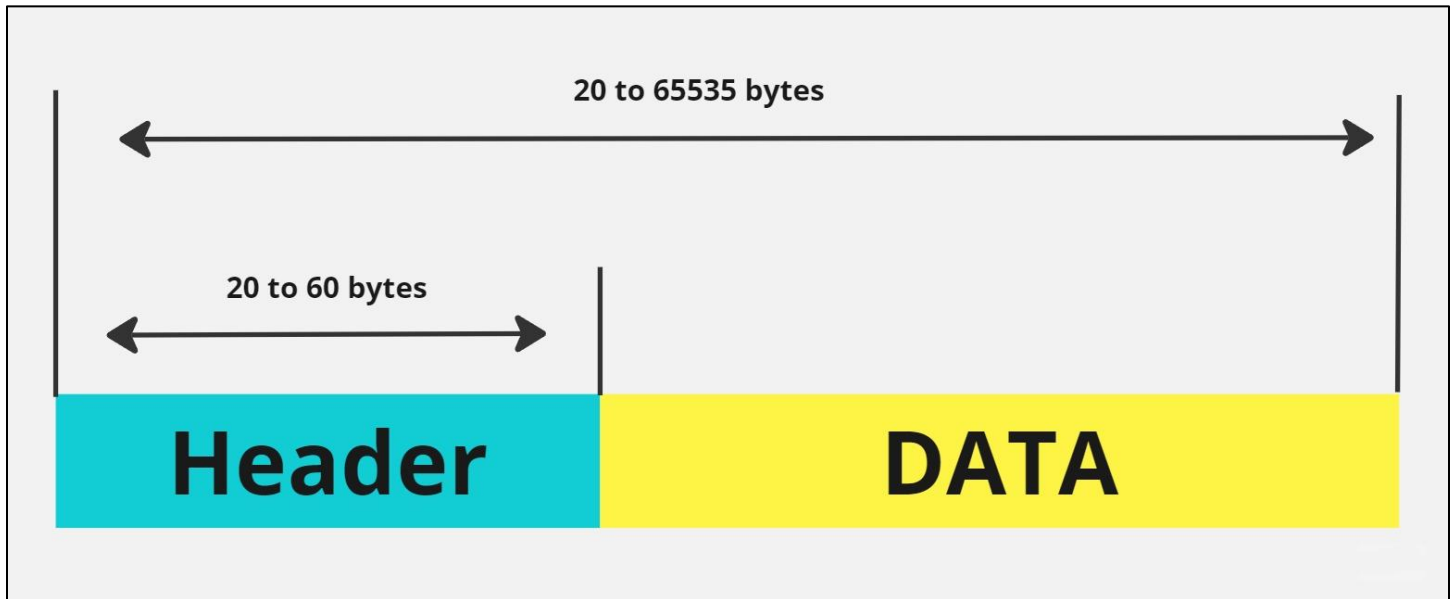
- VLSM stands for Variable Length Subnet Mask.
- VLSM allows us to allocate IPs according to our needs.

We start from the networks with the highest requirement.

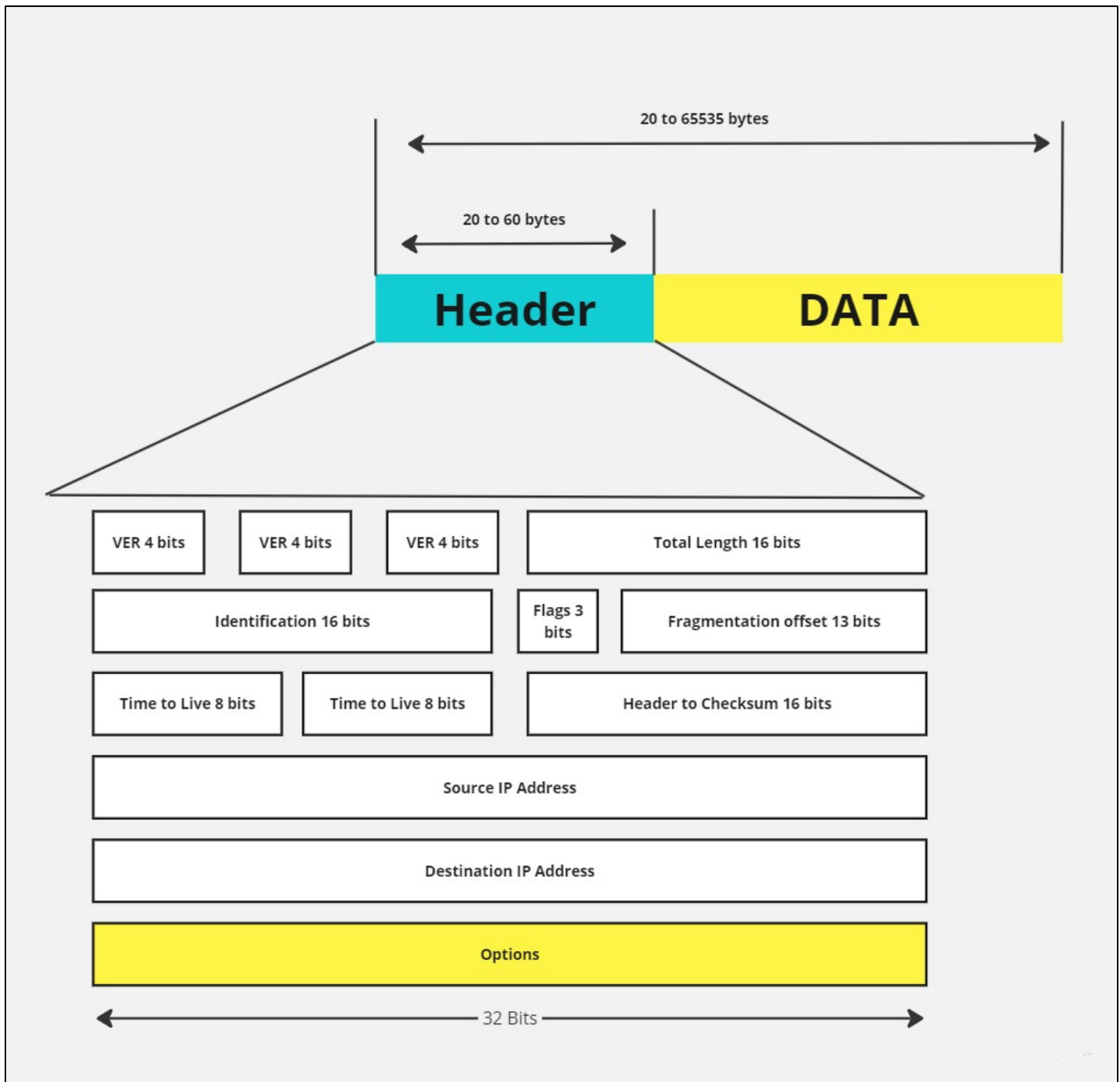
Required IPs	Default Subnet Mask	Network Bits	Host Bits	Usable IPs	Subnet Mask (Bit Count)	Subnet Mask	IP Range
40	255.255.255.0	24	8	62	8 - 6 = 2 (Subnet /26)	255.255.255.192	192.168.1.0 - 192.168.1.63
30	255.255.255.0	24	8	30	8 - 5 = 3 (Subnet /27)	255.255.255.224	192.168.1.64 - 192.168.1.95
20	255.255.255.0	24	8	14	8 - 4 = 4 (Subnet /28)	255.255.255.240	192.168.1.96 - 192.168.1.111

Format of An IPv4 Packets

Format of an IP Version 4: The Header & the Data.



- The size of the **IP header** ranges from **20 to 60 bytes**.
- The header includes the IP routing information that devices use to direct the packet.
- The length of the data section of an IPv4 packet can vary greatly in size. However, the maximum possible size of an IP packet is 65,536 bytes. It contains the message being transferred to the transmission, like website information or email text.



Internet Protocol V6

IPv6 was introduced to address the shortage of IPv4 addresses.

The size of IPv6 addresses is **128 bits**.

IPv4 has four octets (e.g., 192.168.1.0), with each octet consisting of 8 bits, totaling 32 bits.

IPv6 has 8 segments, each part being 16 bits.

Example: **ffff:abcd:shsh:8587:7632:agsv:asuy:8362**

16x8 = 128 bits.

IPv6 is classless.

IPv6 is not an upgraded version of **IPv4**; both are different.

Larger Address

- **Header Size**
 - **IPv4 has 20 bytes.**
 - **IPv6 has 40 bytes.**

128 bits + additional information = **40 bytes**.

IPv4 has 16 information sections to read.

IPv6 has 8 information sections to read.

NAT is not required.

IPv4 had Private & Public IP concepts to address the IP shortage, but IPv6 provides an abundance of IPs.

IPv6 supports **Statefull**, Stateless, and automatic IP generation using MAC addresses (PCs can automatically generate IPs through their own MAC addresses).

IPv4 supports **Unicast, Multicast, Broadcast**, while IPv6 has Unicast, Multicast, Anycast (nearby devices), but no Broadcast.

- Example:
 - Phone -> Airtel Tower -> Provided IP (random)

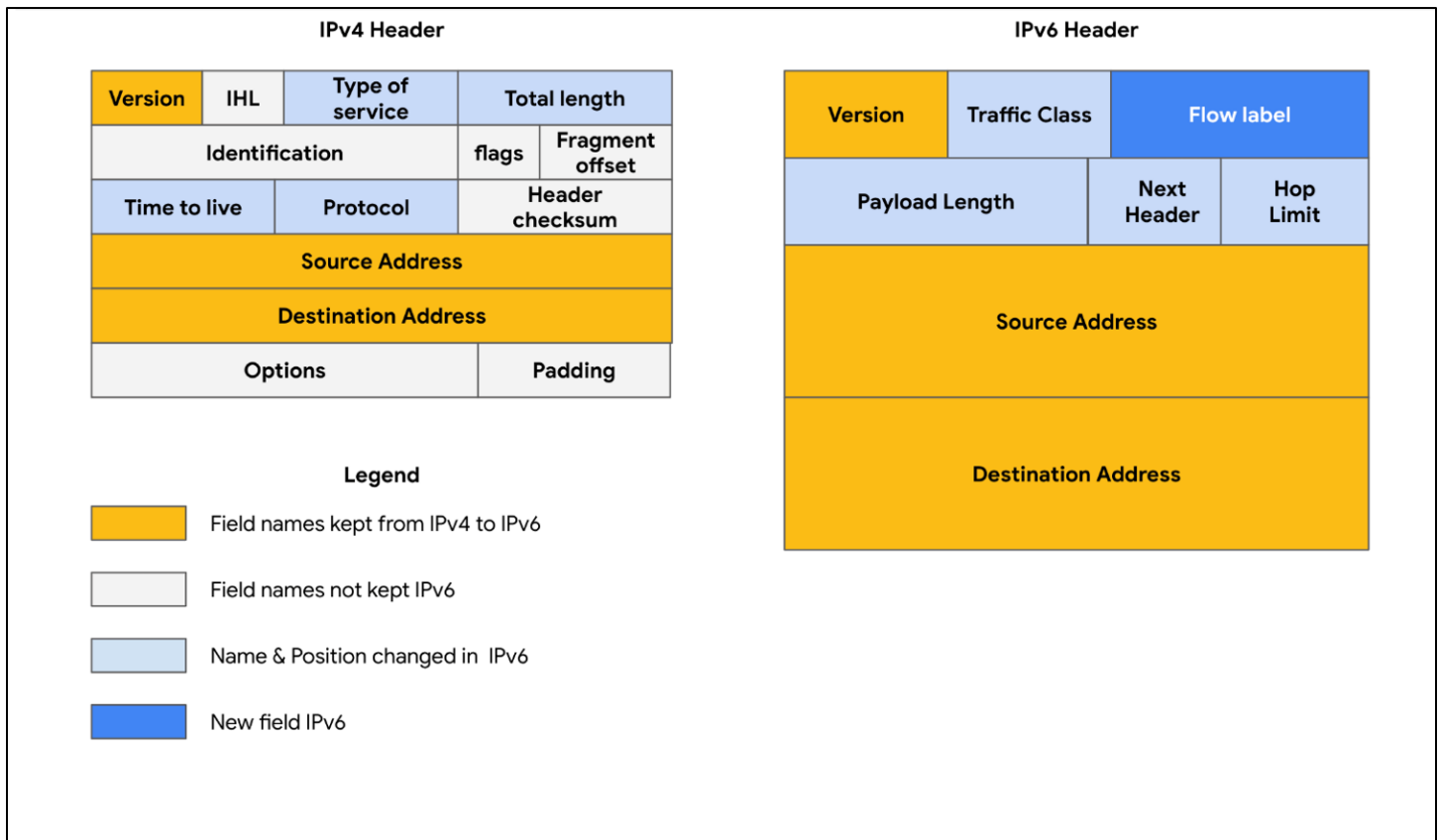
In **IPv6**, even if the location changes, the IP may not change; however, an ISP like Airtel could provide an option to maintain the same IP for payment.

Location changing in **IPv6** doesn't necessarily result in IP changes.

VPN supports **IPv6** features.

Internet Protocol v4 VS v6

One of the key differences between IPv4 and IPv6 is the length of the addresses.
IPv4 addresses are numeric, made of 4 bytes, and allow for up to 4.3 billion possible addresses.
IPv4 addresses are made up of four strings, and the numbers range from 0 to 255.
An example of an IPv4 address would be: 198.51.100.0. IPv6 addresses are hexadecimal, made up of 16 bytes, and allow for up to 340 undecillion addresses (340 followed by 36 zeros).
An example of an IPv6 address would be: 2002:0db8:0000:0000:0000:ff21:0023:1234.
There are also some differences in the layout of an IPv6 packet header . The IPv6 header format is much simpler than IPv4. For example, the IPv4 Header includes the HLEN Identification, and Flags fields, whereas the IPv6 does not. The IPv6 header introduces different fields not included in IPv4 headers, such as the Flow Label and Traffic Class.
There are some important security differences between IPv4 and IPv6.
IPv6 offers more efficient routing and eliminates private address collisions that can occur on IPv4 when two devices on the same network are attempting to use the same address.



OSI MODEL LAYER

The **Open System Interconnection Model** is a standard in Networking which is now known as **TCP/IP**.

International Standard Organization (ISO)

It's an organization that sets international Standards for various measurements.

Open System InterConnection (OSI)

The OSI model describes **Seven Layers** used by endpoint (Computer) system to communicate over a Network.

Its depends in connecting different vendors and PCs through Standardized protocol/Rules.

OSI Model Layers

Layer Number	Layer Name
7	Application Layer
6	Presentation Layer
5	Session Layer
4	Transport Layer
3	Network Layer
2	Data Link Layer
1	Physical Layer

Application Layer

Responsible for making connections and interface with other devices using **HTTP, HTTPS, FTP and more.**

Presentation Layer

Handles the translation, encryption, and compression of data. It ensures that the data sent from the application layer of one system is readable by the application layer of another system.

Session Layer

What is Session ID or Token ?

A User cannot enter credentials every time certain action is performed therefore there server create SESSIONID in USERID database which generate a random string used as a user credentials

Session layer is responsible for managing session, Maintenance and termination of sessions. while handles authentication and reconnection which ensures continuous data flow in a session

Transport Layer

Transport Layer is the heart of OSI Model Its responsible for Communication between NODES (Endpoints to Servers) ensuring the data transfer , speeds and segmentations (Divide Larger data into manageable segments).

- Corrects error from DATA LINK LAYER

- This is where TCP and UDP Protocol Works more into on

Network Layer

Network Layer is responsible for **ROUTING (Data between networks and handles packets forwarding)**. Which means Manages IP Addressing Routing and Firewalling. and converts the data into **PACKETS**

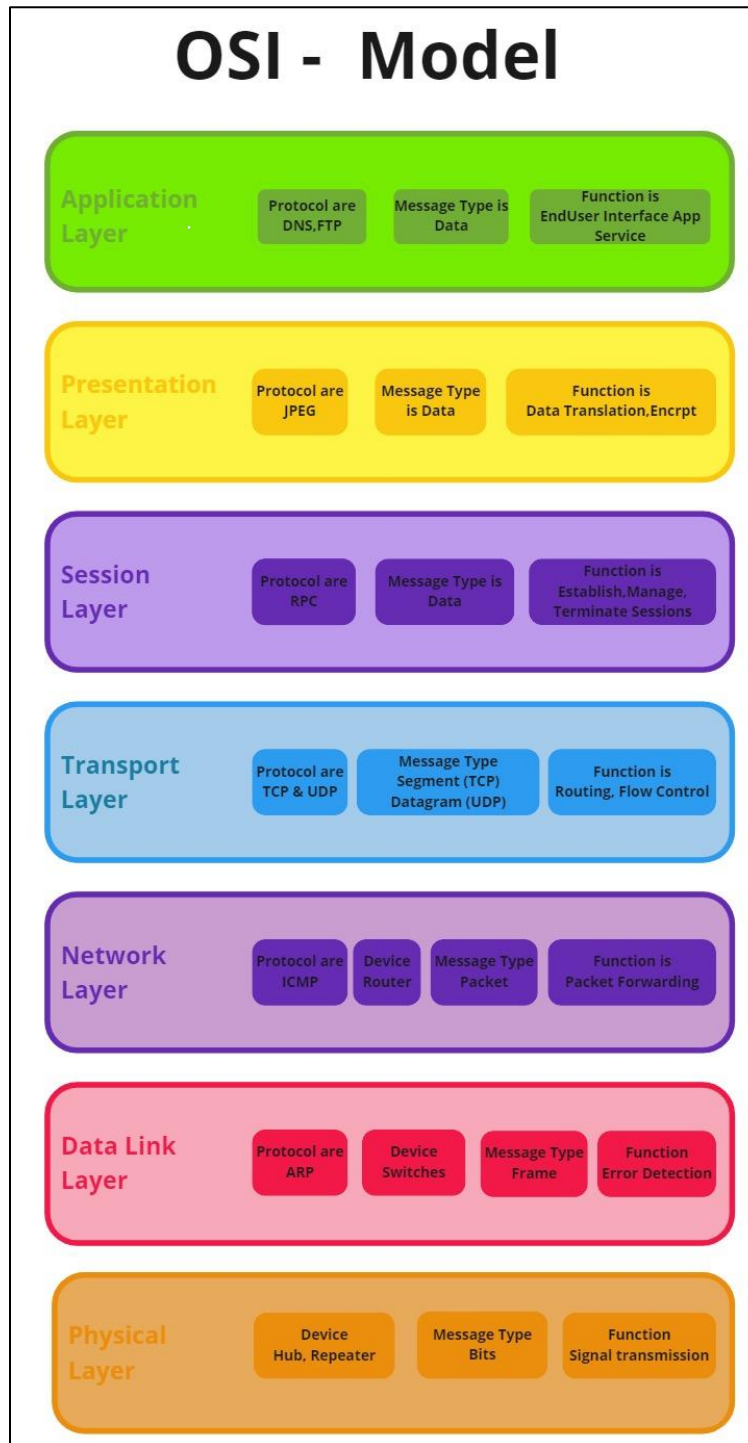
Data Link Layer

Data Link layer ensures reliable data transfer between two directly connected devices over a network. also manages data frames and Error Detection and Error Correction. Involves MAC Address and Network Interface Cards. Lastly convert the **PACKETS** into **FRAMES**

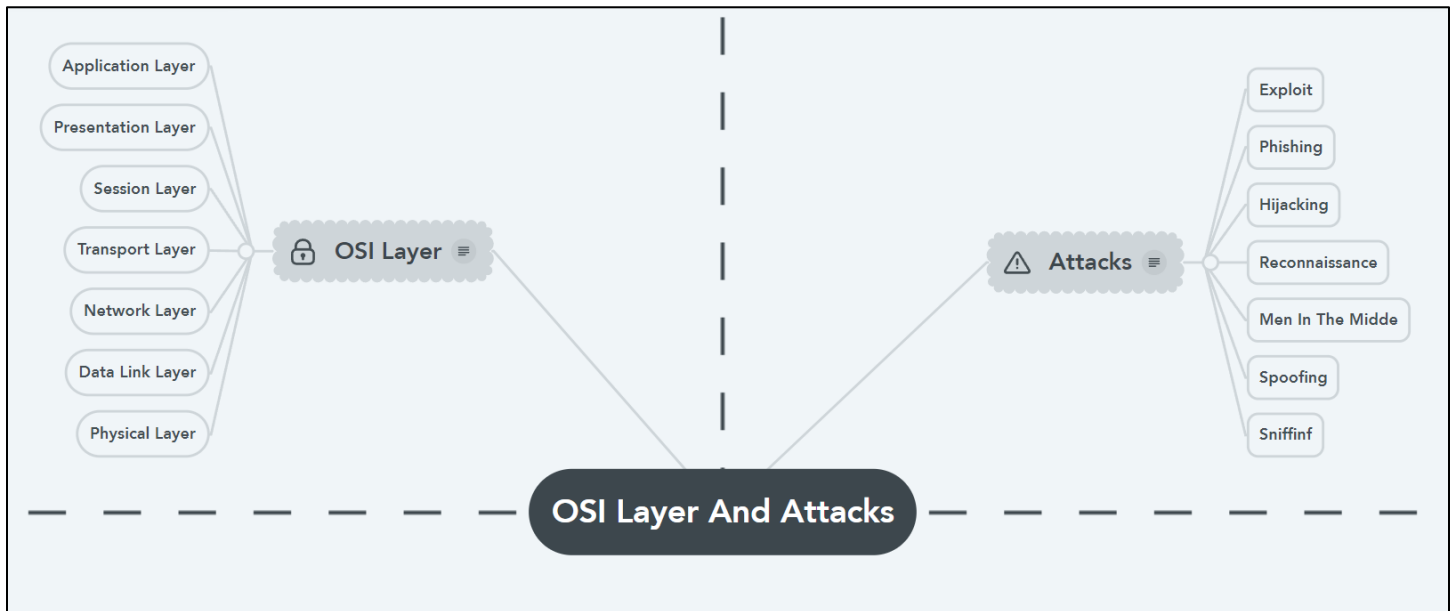
Physical Layer

- Physical Layer is all about physical Network Connectivity (speed and power supply)
- Responsible for the physical transmission of data.
- Involves hardware likes cables, modems.
- Converts the **FRAMES** into **BINARY**

OSI Topology



OSI Layer Vs Attacker Layer



TCP / IP Protocol

TCP/IP stands for **Transmission Control Protocol/Internet Protocol**, and it is used for communication protocols. It combines TCP and IP protocols

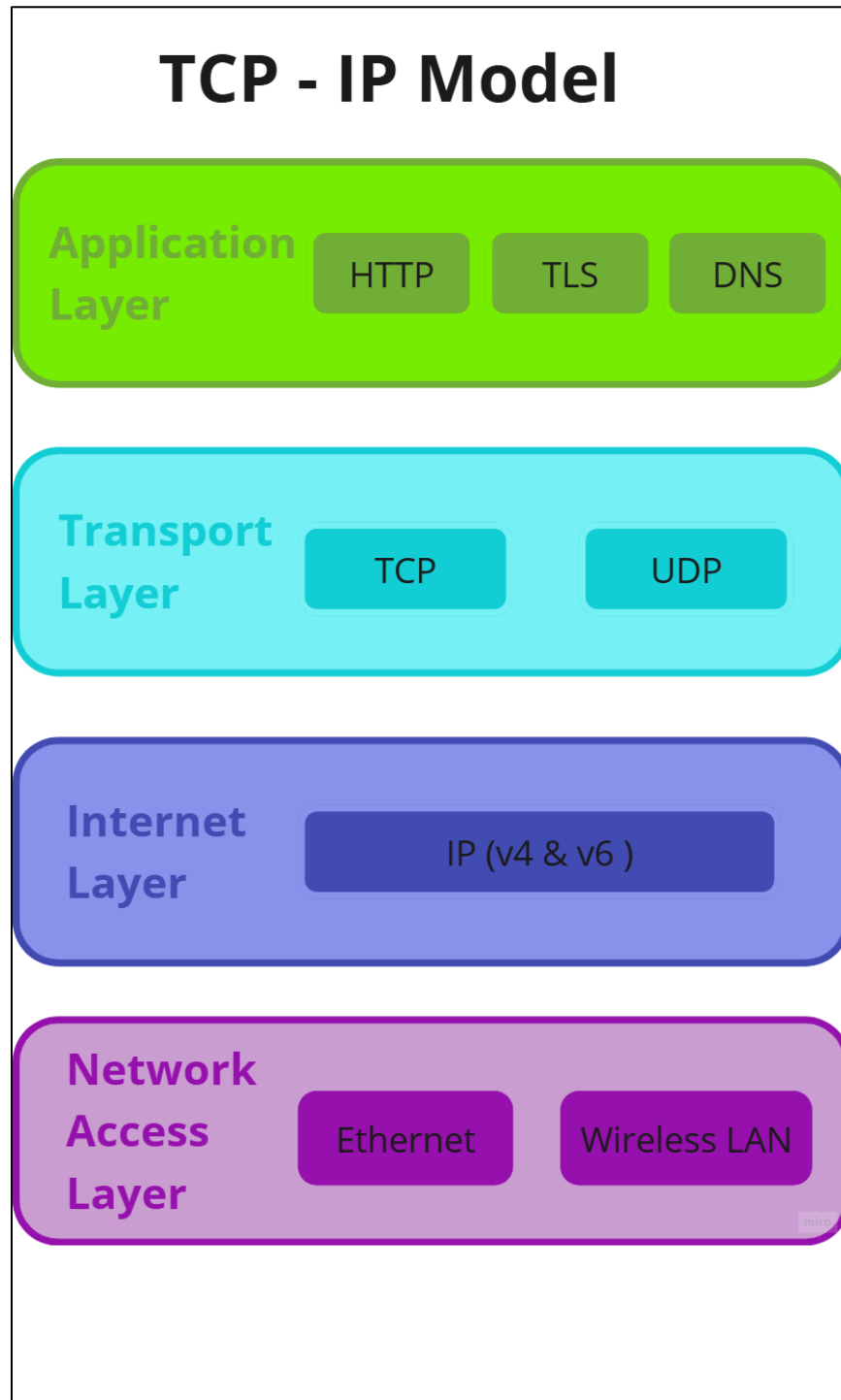
Nowadays QUIC protocol is used which is a combination of TCP , UDP , HTTP.

The **TCP/IP** model provides a framework for understanding how data is organized and transmitted across a network. This model assists network engineers and security analysts in conceptualizing network processes and identifying disruptions or security threats. The TCP/IP model comprises four layers: Network Access Layer, Internet Layer, Transport Layer, and Application Layer. When troubleshooting network issues, security professionals can analyze which layers were involved in an incident.

TCP / IP Layers

Layer Number	Layer Name	Corresponding OSI Layer
1	Application Layer	Application, Presentation, Session Layers
2	Transport Layer	Transport Layer
3	Network Layer	Network Layer
4	Network Interface Layer	Data Link, Physical Layers

TCP/IP Network Topology



Network Protocol And Service

Network protocols are a set of rules that govern how devices communicate across a network